Probability Study Guide

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1. What probability is really trying to do

Probability puts a number between 0 and 1 on how likely something is to happen when the result is uncertain.

- 0 means it cannot happen.
- 1 means it is certain.
- 0.5 means it is a coin-flip.

Think of it as a tool for making decisions under uncertainty: will it rain, will a customer buy, will a machine fail. In data science and machine learning, almost every model is a way of estimating probabilities.

Two ways to think about it:

- **Long-run frequency:** if you flip a fair coin thousands of times, about half come up heads, so P(heads) = 0.5.
- **Degree of belief:** how confident you are about something, such as a 70% chance your team wins.

For exams, the basic idea is enough:

P(event) = (number of favourable outcomes) ÷ (total number of equally likely outcomes)

Example: a fair die is rolled. P(getting a 4) = 1/6, and P(an even number) = 3/6 = 1/2.

2. Basic language of probability

Learn these words first; every later topic uses them.

Term	Meaning	Example
Experiment	Any process with an uncertain result	Rolling a die
Outcome	One single possible result	Getting a 3

Term	Meaning	Example
Sample space (S)	The set of all possible outcomes	{1, 2, 3, 4, 5, 6}
Event	A group of outcomes we care about	Getting an even number = {2, 4, 6}
Trial	One run of the experiment	One roll
Probability P(A)	How likely event A is, from 0 to 1	P(even) = 0.5

Tip: an outcome is one thing that can happen. An event is a collection of outcomes.

3. Events, sample space, and outcomes

Step one of almost every probability problem: write down the sample space, then mark which outcomes belong to your event.

Example: tossing two coins.

- Sample space S = {HH, HT, TH, TT}, so 4 outcomes.
- Event A = "at least one head" = {HH, HT, TH}.
- P(A) = 3/4.

Types of events:

- **Simple event:** one outcome only, such as {HH}.
- **Compound event:** more than one outcome, such as "at least one head".
- **Mutually exclusive events:** cannot happen together, such as rolling a 2 and rolling a 5 on one throw.
- **Exhaustive events:** together they cover the whole sample space.

Tip: if the outcomes are not equally likely, you cannot just count. Use each outcome's own probability instead.

4. Probability rules you must know first

Three rules (axioms) that every probability follows:

1. P(A) is never negative, and never more than 1.
2. P(whole sample space) = 1. Something must happen.

3. For mutually exclusive events, the probabilities add up.

Useful rules that follow from them:

Rule	Formula	Example
Range	$0 \leqslant P(A) \leqslant 1$	P(rain) = 0.3
Complement	P(not A) = 1 − P(A)	P(no rain) = 0.7
Impossible event	$P(\varnothing) = 0$	Rolling a 7 on one die
Addition (exclusive)	P(A or B) = P(A) + P(B)	P(2 or 5) = 1/6 + 1/6 = 1/3

Shortcut: when "at least one" appears in a question, compute 1 − P(none). It is usually much faster.

5. Union, intersection, and complement

These describe how events combine.

Name	Symbol	Plain meaning
Union	$A \cup B$	A or B (or both) happens
Intersection	$A \cap B$	A and B both happen
Complement	A' (or not A)	A does not happen

The general addition rule works for any two events. It subtracts the overlap because the overlap was counted twice:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example: in a class, 60% study maths, 40% study physics, and 25% study both. P(maths or physics) = 0.60 + 0.40 − 0.25 = 0.75. So P(neither) = 1 − 0.75 = 0.25.

If A and B are mutually exclusive, $P(A \cap B) = 0$ and the rule reduces to simple addition.

6. Conditional probability

Conditional probability is the chance of A once you already know B has happened. It is written P(A | B), read "probability of A given B". Knowing B shrinks the sample space to just B.

P(A | B) = P(A ∩ B) ÷ P(B), where P(B) > 0

Example: a card is drawn from a standard deck. Given that it is a face card (J, Q, K: 12 cards), what is the chance it is a king?

- P(king and face card) = 4/52
- P(face card) = 12/52
- P(king | face) = (4/52) / (12/52) = 1/3

Rearranging gives the multiplication rule, which is useful for sequences of events:

P(A ∩ B) = P(A | B) × P(B)

Tip: P(A | B) and P(B | A) are usually different. Always check which event is the "given" one.

7. Independence vs dependence

Two events are **independent** if one happening does not change the chance of the other. Otherwise they are **dependent**.

Independent events: P(A ∩ B) = P(A) × P(B), which is the same as P(A | B) = P(A)

	Independent	Dependent
Meaning	One event does not affect the other	One event changes the other's chance
Example	Two separate coin tosses	Drawing two cards without replacement
P(A and B)	P(A) × P(B)	P(A) × P(B given A)

Example (dependent): a bag has 3 red and 2 blue balls. Two are drawn without replacement. P(both red) = 3/5 × 2/4 = 3/10.

Example (independent): with replacement, the same draw gives 3/5 × 3/5 = 9/25.

Common trap: independent is not the same as mutually exclusive. Mutually exclusive events with non-zero probabilities are always dependent, because if one happens the other cannot.

8. Bayes' theorem

Bayes' theorem flips a conditional probability. You know P(B | A) and want P(A | B), for example going from "test result given disease" to "disease given test result".

P(A | B) = [P(B | A) × P(A)] ÷ P(B)

The denominator P(B) is usually found with the Law of Total Probability (next section).

Example: medical test. 1% of people have a disease. The test catches 95% of sick people (true positive) and wrongly flags 5% of healthy people (false positive). You test positive. What is the chance you are actually sick?

- P(positive) = 0.95 × 0.01 + 0.05 × 0.99 = 0.0095 + 0.0495 = 0.059
- P(sick | positive) = 0.0095 / 0.059 ≈ 0.161

Only about 16%, because the disease is rare and most positives come from the large healthy group.

Vocabulary: P(A) is the prior (belief before the evidence), P(A | B) is the posterior (belief after the evidence).

9. Law of Total Probability

When the world splits into separate cases (B_1, B_2, …, B_n) that do not overlap and cover everything, the chance of A is the weighted sum across the cases.

$P(A) = P(A \mid B_1)\cdot P(B_1) + P(A \mid B_2)\cdot P(B_2) + \ldots + P(A \mid B_n)\cdot P(B_n)$

Example: a factory has two machines. Machine 1 makes 60% of items with a 2% defect rate. Machine 2 makes 40% with a 5% defect rate.

P(defective) = 0.02 × 0.60 + 0.05 × 0.40 = 0.012 + 0.020 = 0.032 (3.2%).

When to use it: whenever the question says "overall" or "in total" and the data is given by group.

10. Random variables and probability distributions

A **random variable** (usually X) is a number that depends on the result of a random experiment. It turns outcomes into numbers so we can do maths on them.

Example: toss two coins and let X = number of heads. X can be 0, 1, or 2.

A **probability distribution** lists every value X can take and how likely each one is.

X (heads)	0	1	2
P(X = x)	0.25	0.50	0.25

The probabilities always add up to 1.

Two key summaries:

$E(X) = \Sigma\, x \cdot P(x)$ and **$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$**

- **Expected value E(X)** is the long-run average. Here: 0(0.25) + 1(0.5) + 2(0.25) = 1.
- **Variance** measures spread. Standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$.

11. Discrete vs continuous distributions

	Discrete	Continuous
Values	Countable, often whole numbers	Any value in a range
Typical question	How many?	How much? How long?
Examples	Number of heads, defects, calls	Height, weight, waiting time
P(X = exact value)	Can be positive	Always 0
Probability comes from	Adding up values	Area under a curve
Function used	PMF	PDF
Examples of distributions	Bernoulli, Binomial, Poisson	Uniform, Normal

Why P(X = exact value) is 0 for continuous: there are infinitely many possible values, so any single point has zero width. We ask about intervals instead, such as P(160 < height < 170).

Quick test: if you count it, it is discrete. If you measure it, it is continuous.

12. Probability mass function, density function, and cumulative distribution

PMF (discrete). Gives P(X = x) for each value. Rules: every value is between 0 and 1, and all values sum to 1.

PDF (continuous). A curve f(x). The probability of an interval is the area under the curve. Rules: f(x) ⩾ 0, and the total area is 1. The height of a PDF is not a probability.

P(a ⩽ X ⩽ b) = area under the curve f(x) between a and b

CDF (both types). The chance that X is at most x. It only ever goes up, from 0 to 1.

F(x) = P(X ⩽ x)

Task	Discrete	Continuous
P(X ⩽ x)	Add PMF values up to x	F(x), or area up to x
P(a < X ⩽ b)	F(b) − F(a)	F(b) − F(a)
P(X > x)	1 − F(x)	1 − F(x)

Tip: the CDF is the easiest route to "at most", "at least" and "between" questions.

13. Bernoulli distribution

The simplest distribution: **one trial with two outcomes**, success (1) or failure (0). Success has probability p.

P(X = 1) = p and **P(X = 0) = 1 − p**

- Mean: E(X) = p
- Variance: p(1 − p)

Examples: one coin toss, one customer buys or not, one email is spam or not.

Example: a free throw succeeds with p = 0.7. Then P(miss) = 0.3, mean = 0.7, variance = 0.7 × 0.3 = 0.21.

This is the building block for the Binomial distribution.

14. Binomial distribution

Count the number of successes in **n independent Bernoulli trials**, each with the same success probability p. Write X ~ Bin(n, p).

Use it when:

- There is a fixed number of trials n.
- Each trial is success or failure.
- p is the same every time.
- Trials are independent.

$P(X = k) = C(n, k) \times p^k \times (1 - p)^{n-k}$, for k = 0, 1, 2, …, n

Here C(n, k) = n! / (k!(n − k)!) counts the ways to choose which k trials succeed.

- Mean: np
- Variance: np(1 − p)

Example: a coin is tossed 5 times. P(exactly 3 heads) = $C(5,3) \times 0.5^3 \times 0.5^2 = 10 \times 0.03125 = 0.3125$.

Example: 10 items are tested, each defective with p = 0.1. Expected defects = 10 × 0.1 = 1.

15. Poisson distribution

Counts how many times an event happens in a **fixed interval of time or space**, when events occur independently at a steady average rate λ (lambda). Write X ~ Poisson(λ).

$P(X = k) = (e^{-\lambda} \times \lambda^k) \div k!$, for k = 0, 1, 2, …

- Mean = λ
- Variance = λ (mean and variance are equal)

Examples: calls per hour at a help desk, typos per page, accidents per week.

Example: a shop averages 3 customers per 10 minutes. P(exactly 2 in the next 10 minutes) = $e^{-3} \times 3^2 / 2! = 0.0498 \times 4.5 \approx 0.224$.

Tip: if the interval changes, scale λ. 3 per 10 minutes means λ = 6 for 20 minutes.

Link to Binomial: when n is large and p is small, Bin(n, p) is well approximated by Poisson with λ = np.

16. Uniform distribution

Every value in an interval [a, b] is **equally likely**. The PDF is a flat line. Write X ~ U(a, b).

f(x) = 1 ÷ (b − a) for a ⩽ x ⩽ b, and **P(c ⩽ X ⩽ d) = (d − c) ÷ (b − a)**

- Mean: (a + b) / 2
- Variance: (b − a)² / 12

Example: a bus arrives at a random time within a 20-minute window, so X ~ U(0, 20). P(wait under 5 minutes) = 5/20 = 0.25. Mean wait = 10 minutes.

Tip: probability is just the fraction of the interval you are asking about.

17. Normal distribution

The famous bell curve. It is symmetric around the mean μ, and its width is set by the standard deviation σ. Write X ~ N(μ, σ²). Heights, test scores and measurement errors are often close to normal.

f(x) = [1 ÷ (σ√(2π))] × e^(−(x − μ)² ÷ (2σ²))

The 68–95–99.7 rule:

Range	Share of data
μ ± 1σ	about 68%
μ ± 2σ	about 95%
μ ± 3σ	about 99.7%

Standardising (z-score). To find probabilities, convert X to the standard normal Z, which has mean 0 and standard deviation 1, then use a Z table or calculator.

Z = (X − μ) ÷ σ

Example: exam scores ~ N(60, 10²). P(X > 80)?

- z = (80 − 60) / 10 = 2
- P(Z > 2) = 1 − 0.9772 = 0.0228, about 2.3%

Useful facts: the curve is symmetric, so P(Z < −z) = P(Z > z). Mean = median = mode.

18. Comparing the important distributions

Pick the distribution by asking what you are counting or measuring.

Distribution	Type	Models	Parameters	Mean	Variance
Bernoulli	Discrete	One yes/no trial	p	p	$p(1-p)$
Binomial	Discrete	Successes in n trials	n, p	np	$np(1-p)$
Poisson	Discrete	Events in a fixed interval	λ	λ	λ
Uniform	Continuous	Equally likely values in $[a, b]$	a, b	$(a+b)/2$	$(b-a)^2/12$
Normal	Continuous	Bell-shaped measurements	μ, σ	μ	σ^2

Quick chooser:

- One trial, yes or no → Bernoulli.
- Fixed number of trials, count successes → Binomial.
- Count of events over time or space, no fixed upper limit → Poisson.
- All values in a range equally likely → Uniform.
- Measurement that clusters around an average → Normal.

19. How to solve typical exam-style questions

Follow the same five steps each time:

1. **Read and name the quantity.** What exactly is being asked: a single probability, an "at least", a conditional, a mean?
2. **Define events or the random variable** with letters, such as A = "item is defective" or X = "number of calls".
3. **Identify the type.** Counting outcomes, conditional, Bayes, or a named distribution? Use the quick chooser in section 18.
4. **Write the formula, then substitute.** Showing the formula earns marks even if the arithmetic slips.
5. **Sanity-check.** The answer must lie between 0 and 1, and probabilities of all cases must sum to 1.

Keyword guide:

If the question says	Think
"given that", "if it is known"	Conditional probability
"or"	Add, minus overlap
"and"	Multiply (check independence)
"at least one"	1 − P(none)
"overall", "in total" across groups	Law of Total Probability
"which source / cause" after seeing a result	Bayes' theorem
"n trials, probability of k successes"	Binomial
"average rate per hour/page"	Poisson
"between a and b" for a measurement, mean and sd given	Normal with z-score

Worked example: a test has 10 multiple-choice questions, each guessed with p = 0.25. P(at least 1 correct) = 1 − P(none) = $1 - 0.75^{10} = 1 - 0.0563 \approx 0.944$.

20. Common mistakes and how to avoid them

Mistake	Fix
Adding P(A) + P(B) when events overlap	Subtract P(A ∩ B), or check they are mutually exclusive
Multiplying probabilities for dependent events	Use P(A) × P(B | A)
Mixing up P(A | B) and P(B | A)	Ask which event is the "given" one; use Bayes to flip
Treating "independent" and "mutually exclusive" as the same	Independent: P(A ∩ B) = P(A)P(B). Exclusive: P(A ∩ B) = 0
Forgetting the denominator P(B) in Bayes	Compute it with the Law of Total Probability
Using Binomial when trials are not independent or p changes	Check the four Binomial conditions

Mistake	Fix
Wrong λ in Poisson	Scale λ to match the interval in the question
Treating a PDF height as a probability	Probability is the area under the curve
Using σ instead of σ² (or the reverse) in Normal	$N(\mu, \sigma^2)$ is written with the variance; the z-score uses σ
Answers above 1 or negative	Stop and recheck; probabilities live in [0, 1]
Computing "at least" the long way	Use the complement: 1 − P(none)

21. Final summary sheet

Core rules

- $0 \leqslant P(A) \leqslant 1$, and $P(\text{not } A) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \mid B) = P(A \cap B) \div P(B)$, and $P(A \cap B) = P(A \mid B) \times P(B)$
- Independent events: $P(A \cap B) = P(A) \times P(B)$
- Total probability: $P(A) = \Sigma\, P(A \mid B_i) \times P(B_i)$
- Bayes: $P(A \mid B) = [\, P(B \mid A) \times P(A) \,] \div P(B)$

Random variables

- $E(X) = \Sigma\, x \cdot P(x)$
- $\text{Var}(X) = E(X^2) - [E(X)]^2$
- $F(x) = P(X \leqslant x)$

Distributions at a glance

Distribution	Formula	Mean	Variance
Bernoulli(p)	$P(X=1) = p$	p	$p(1-p)$
Binomial(n, p)	$C(n,k)\, p^k\, (1-p)^{n-k}$	np	$np(1-p)$
Poisson(λ)	$e^{-\lambda}\, \lambda^k / k!$	λ	λ
Uniform(a, b)	$1 / (b-a)$	$(a+b)/2$	$(b-a)^2/12$
Normal(μ, σ²)	$z = (x-\mu)/\sigma$	μ	σ^2

Remember

- Probabilities are between 0 and 1.
- "At least one" means 1 − P(none).
- Discrete → PMF and sums. Continuous → PDF and areas.
- Normal: 68–95–99.7 rule, and always standardise with z.
- Poisson mean equals variance.
- Check your answer is sensible before moving on.